H-CARD: Teaching Team

Prof Etienne Burdet
Course Leader

Human motor control & human-machine interaction

Yihan Liu
GTA

Robotics, mechanics, control

Jingwen Zhao
GTA

Mechatronics, soft robotics

Josh Francis
GTA

Sensing, mechatronics

Paschal Egan
Electronics Laboratories

Electronics engineering

EXAMPLES OF INTERFACES

Movement augmentation

Soft Sixth Finger

CHI 2021

BackTrack: 2D Back-of-device Interaction Through Front Touchscreen

Chang Xiao[1] Karl Bayer[2] Changxi Zheng[1] Shree K. Nayar[2]

[1]Columbia University [2]Snap Inc.

BackTrack

Modular prosthetics for task-specific tools

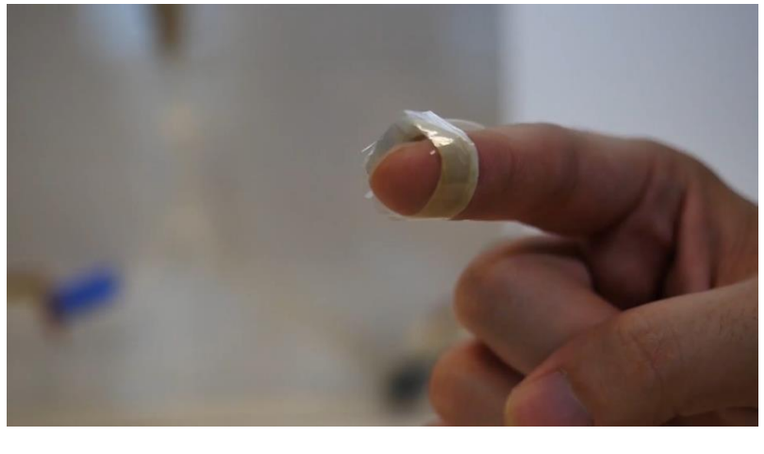
Foot interface for robot operation

Imperial College London

NANYANG TECHNOLOGICAL UNIVERSITY SINGAPORE

Sensory augmentation

Haptic vest



HydroRing

WEEKS 1-3: 16/1-30/1

- 6 lectures
- networking: 1h on 23/1 and 2h on 30/1
- 2 lab inductions (required to get access to the lab) on 30/1

Deliverables:

- form group, create a Teams chat for communication with the teaching team (format “HCARD group_name”) and a Github repository for your group's presentation and code
- report on one of the lectures (individual) – 2/2
- report on design idea (individual) – 5/2

GROUP FORMATION (BY 2/2/2025)

5
Groups

5-6
Students per group

- 27 registrations
- To learn collaborating with colleagues with different background, each group must have:
  4 MSc Human Robotics
  1 BEng/MEng Bioengineering
  1 Aero/NUS
- Please form your group and notice us by 17h, on 2/2/2025 through filling the form shown in the Teams

REPORT ON ONE LECTURE (2/2)

- individual assignment, <1001 words (+ figures and references)
- your report should summarise or extend the topics covered in one of the six lectures (during weeks 1-2)
- your report should be structured as follows:
  - context/background
  - description of the concepts and ideas
  - applications
  - limitations and future work in the field
- your report will be graded on the following criteria: clarity, accuracy, your critical analysis of the idea

REPORT ON 1 DESIGN IDEA (5/2)

- individual assignment, <1001 words (+ figures and references)
- describe a personal design idea for an augmentation interface
- your idea can be (will generally be) different from the idea your group will ultimately select
- your report may describe (but is not limited to):
  - your design idea and why you choose it
  - which function it augments and how it does it
  - how you would develop the interface from the idea

6/2: BRAINSTORMING ON PROJECT IDEAS

- 30’ for each group for presentation and discussion
- prepare a 10-15’ presentation of 1-3 concepts to help your colleagues and us understand your design ideas
- this will not be graded, is to develop ideas
- you must attend the entire session and can discuss other groups’ ideas

PARTS AND BUDGET

- material available in the electronic lab
- list the board and sensors your group will need and give it to the teaching team
- if you need extra amount of board and sensors, please discuss with the teaching team
- each group can use maximally 100£ (in addition to the ‘starter-kit’)
- Josh will give more details about sensors, parts and how to order them

WEEKS 5-9, 13/2-13/3

Now is the time to develop your innovative system!

We will try our best to assist you:

- physical meeting
  - every Friday morning 9–12h
  - send a few bullet points to the teaching team the Thursday before: summary of your project progress and questions/topics you want to discuss
- lab work
- on March 13, you are expected to show us a functional device

COMMUNICATION

- during the class time
- using the HCARD Teams groupchat
- Github repository
- refer to the documents in blackboard (BB)
- announcements will be made on BB
- assignment submission through BB

FINAL COMPETITION (20/3)

- For the demo day, you will have
  - 5’ presentation (with maximally 5 slides)
  - live demonstration of the device/system
- Criteria for your augmentation/assistive system:
  - usefulness
  - innovation
  - design quality
  - how well it works
- Graded by teaching team, invited visitors and your classmates

GROUP REPORT (27/3)

- report on the developed system
- group assignment < 2500 words, including tables (+ figures and references) and unlimited appendix
- it may include some of the following components:

  - ideas and concepts selection
  - design objectives and specifications
  - system working principle
  - device development
  - prototypes and evaluation
  - limitations and future work

INDIVIDUAL ORAL EXAM (~8/5/2025 ?)

- You will be asked about your group project and specifically your contribution in the project
- This is a Q&A format between the student and the teaching team, *no presentation* but you can bring a copy of your report
- Around 15’ per student

GRADING

- 20% individual report on 1 lecture, and on 1 design idea
- 40% competition in group
- 10% group report on the developed system
- 30% individual oral examination on the developed system

i.e., 50% group mark, 50% individual mark

IMPORTANT DATES

- 2/2 @ 17h: deadline for individual report on one lecture
- 2/2 @ 17h: deadline to communicate your group formation
- 5/2 @ 17h: deadline for individual report on a design idea
- 6/2 @ 9–12h: group presentation on design ideas (not graded)
- 20/3@ 9–12h: final competition
- 27/3@ 17h: deadline for final group report
- May 2025: individual oral exam